

Solving the Tower of Hanoi

The *Tower of Hanoi* is a puzzle involving a tower of stacked disks. The puzzle begins with the largest disk on the bottom, and the disk sizes decrease going up. Each disk has a hole in its center so that the disks can be stacked on top of one another on a pole. [Figure 3-4](#) shows a wooden Tower of Hanoi puzzle.



Figure 3-4: A wooden Tower of Hanoi puzzle set

To solve the puzzle, the player must move the stack of disks from one pole to another while following three rules:

- The player can move only one disk at a time.
- The player can move disks only to and from the top of a tower.
- The player can never place a larger disk on top of a smaller disk.

Python's built-in `turtledemo` module has a Tower of Hanoi demonstration that you can see by running `python -m turtledemo` on Windows or `python3 -m turtledemo` on macOS/Linux, and then selecting **minimum_hanoi** from the